

Valid candidate set  
 $U$

Compute  $v_j$  for  
each candidate

Select Top- $q$   
candidates  $K$

For each  $j$  in  $K$ :  
add  $(\mu_j, +1)$

Run SAM2 once and  
record score  $s^{\wedge}(j)$

Pick  $j^*$  with highest  
 $s^{\wedge}(j)$

Keep  $M_t$  from best  
candidate